

Basic Principles of Medicine 2

Module: Endocrine and Reproductive System| Lecture 14

Clinical Anatomy and Imaging of the Reproductive System - Female Reproductive Organs

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Objectives

SOM.MKII.BPM2.1.ER.1.ANAT.1020	Identify the major female pelvic organs, including the bladder, uterus, uterine tubes, ovaries, vaginal canal and rectum
SOM.MKII.BPM2.1.ER.1.ANAT.1119	Describe the anatomical relationships of the female pelvic organs to each other and surrounding structures, including the pelvic diaphragm and the bony pelvis
SOM.MKII.BPM2.1.ER.1.ANAT.1131	Identify the female pelvic organs on medical imaging studies such as MRI, CT scans, cystourethrography, hysterosalpingogram etc
SOM.MKII.BPM2.1.ER.1.ANAT.1022	Describe the ovaries and ovarian ligaments.
SOM.MKII.BPM2.1.ER.1.ANAT.1023	Describe the parts of the uterus and uterine tubes.
SOM.MKII.BPM2.1.ER.1.ANAT.1024	Define the terms anteverted, anteflexed, retroverted and retroflexed with respect to the position of the uterus.
SOM.MKII.BPM2.1.ER.1.ANAT.1026	Describe the broad ligament of the uterus and its subdivisions in relationship to the body and fundus of the uterus, the uterine tubes and their fimbriated ends, and the round ligament of the uterus.
SOM.MKII.BPM2.1.ER.1.ANAT.1027	Explain the clinical significance of the transverse cervical (cardinal), pubocervical and uterosacral ligaments in uterine support.
SOM.MKII.BPM2.1.ER.1.ANAT.1029	Identify the uterus and uterine tubes on a hysterosalpingogram and its clinical significance
SOM.MKII.BPM2.1.ER.1.ANAT.1021	Describe the reflections of the peritoneum over the bladder, uterus, uterine tubes, ovaries, vaginal canal and rectum in the pelvic cavity.
SOM.MKII.BPM2.1.ER.1.ANAT.1025	Describe the typical relationship of the cervix to the vaginal canal.
SOM.MKII.BPM2.1.ER.1.ANAT.1031	Describe the parts and relationships of the vagina.
SOM.MKII.BPM2.1.ER.1.ANAT.1037	Explain the clinical importance of the relationship of the posterior fornix to the rectouterine pouch.
SOM.MKII.BPM2.1.ER.1.ANAT.1033	Describe the blood supply, venous and lymphatic drainage of the ovaries, uterine tubes, fundus, body and cervix of uterus, pelvic part of vagina, vaginal vestibule.
SOM.MKII.BPM2.1.ER.1.ANAT.1034	Describe the course of the uterine arteries across the pelvic floor.
SOM.MKII.BPM2.1.ER.1.ANAT.1035	Describe “the water under the bridge” relationship and use this information to discuss surgical / anatomical considerations during hysterectomy.
SOM.MKII.BPM2.1.ER.1.ANAT.1127	Explain how fluid in the abdomen can reach the female pelvic peritoneal pouches and where it would ultimately settle.
SOM.MKII.BPM2.1.ER.1.ANAT.1121	Identify the female pelvic organs on medical imaging studies such as MRI, CT scans, etc.
SOM.MKII.BPM2.1.ER.1.ANAT.1028	Define the term “uterine prolapse”, list the structures that are compromised and explain why it is more common in multiparous women who delivered via the vaginal canal.
SOM.MKII.BPM2.1.ER.1.ANAT.1032	Discuss the importance of vaginal and rectal examinations and “pap smears”.
SOM.MKII.BPM2.1.ER.1.ANAT.1030	Explain how lower genitourinary infections in females, such as gonorrhea and chlamydia, can gain access to the pelvic peritoneal spaces.

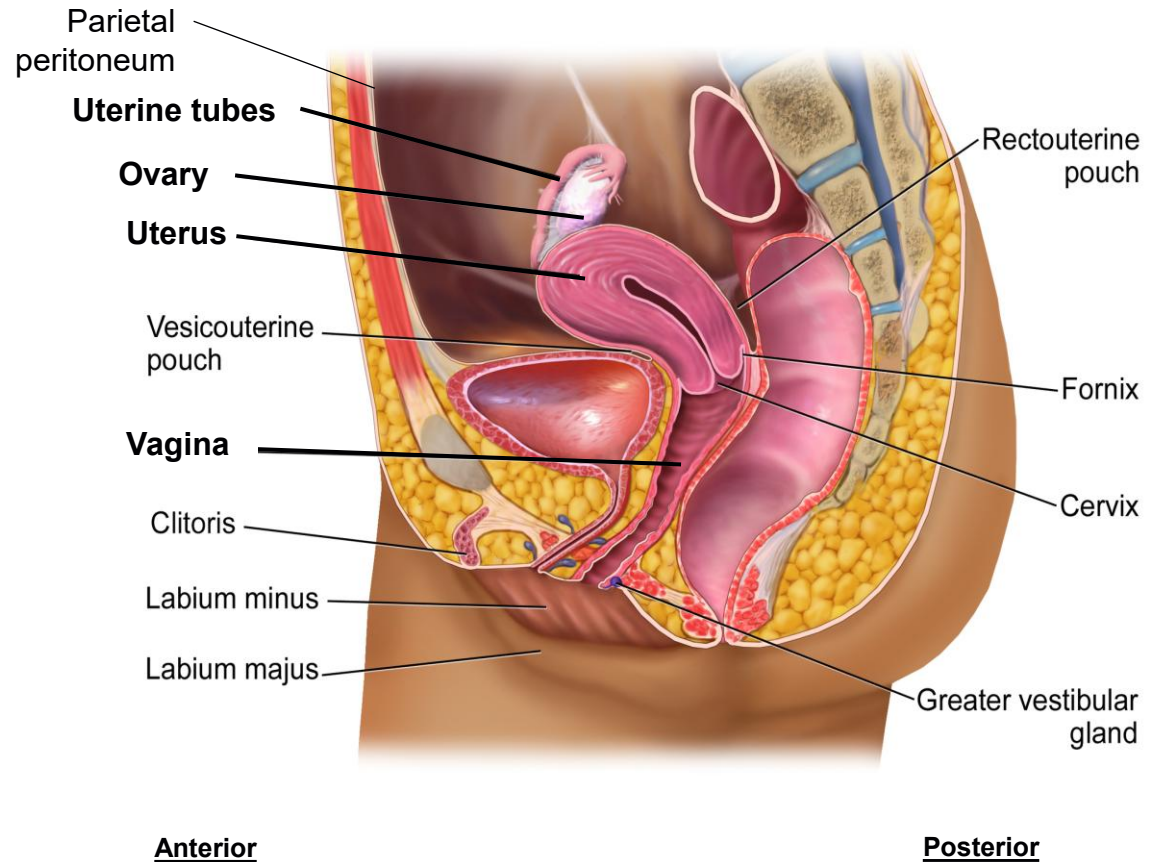
Overview

- Relationship of female pelvic organs
- Clinical anatomy of the female reproductive organs
- Blood supply and venous drainage of female reproductive organs
- General lymphatic drainage of female reproductive organs
- Imaging

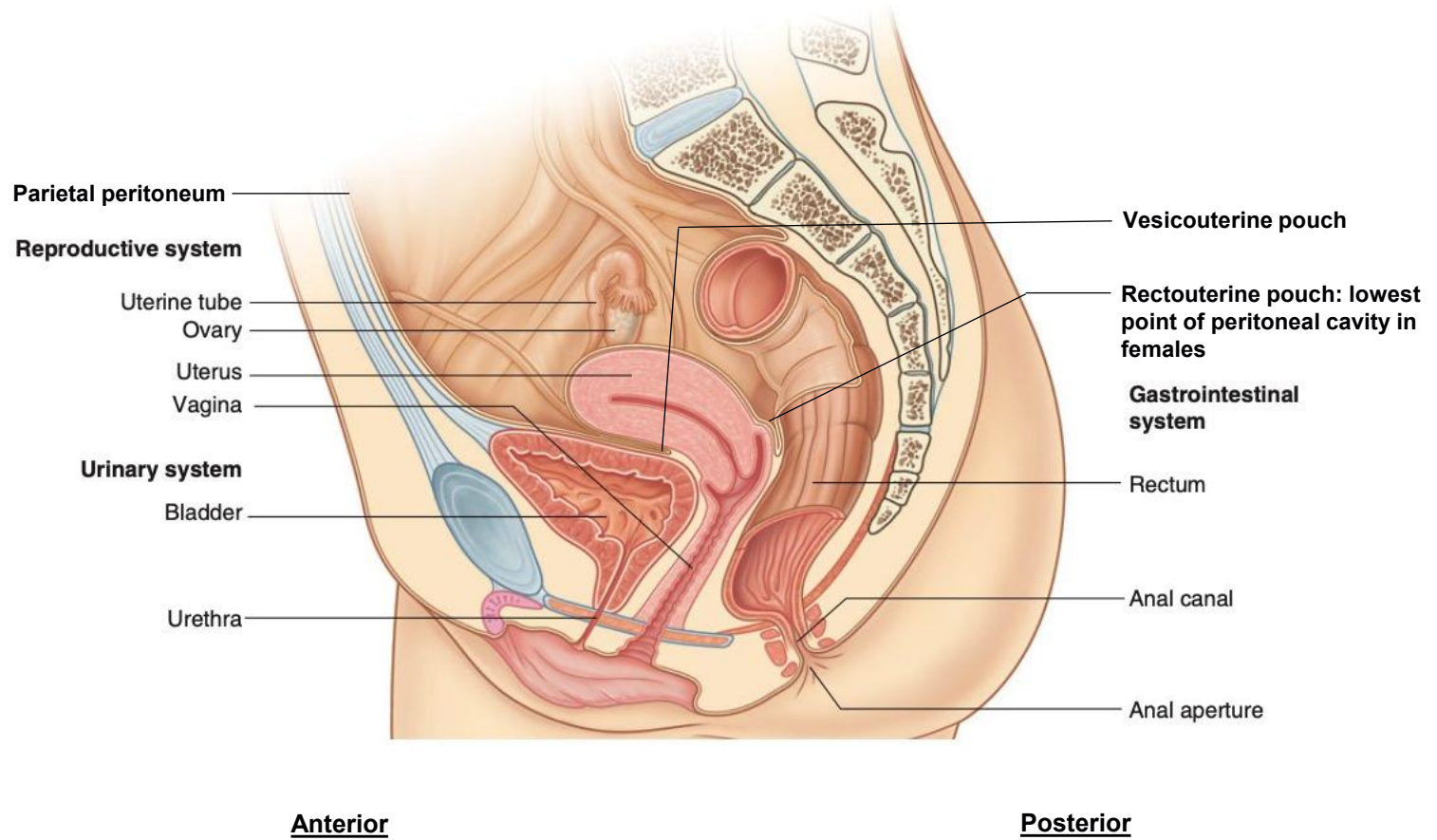
Female reproductive organs

The female reproductive organs include:

- Uterus – unpaired (becomes an abdominopelvic organ during pregnancy)
- Uterine tubes - paired
- Ovaries - paired
- Vagina - unpaired



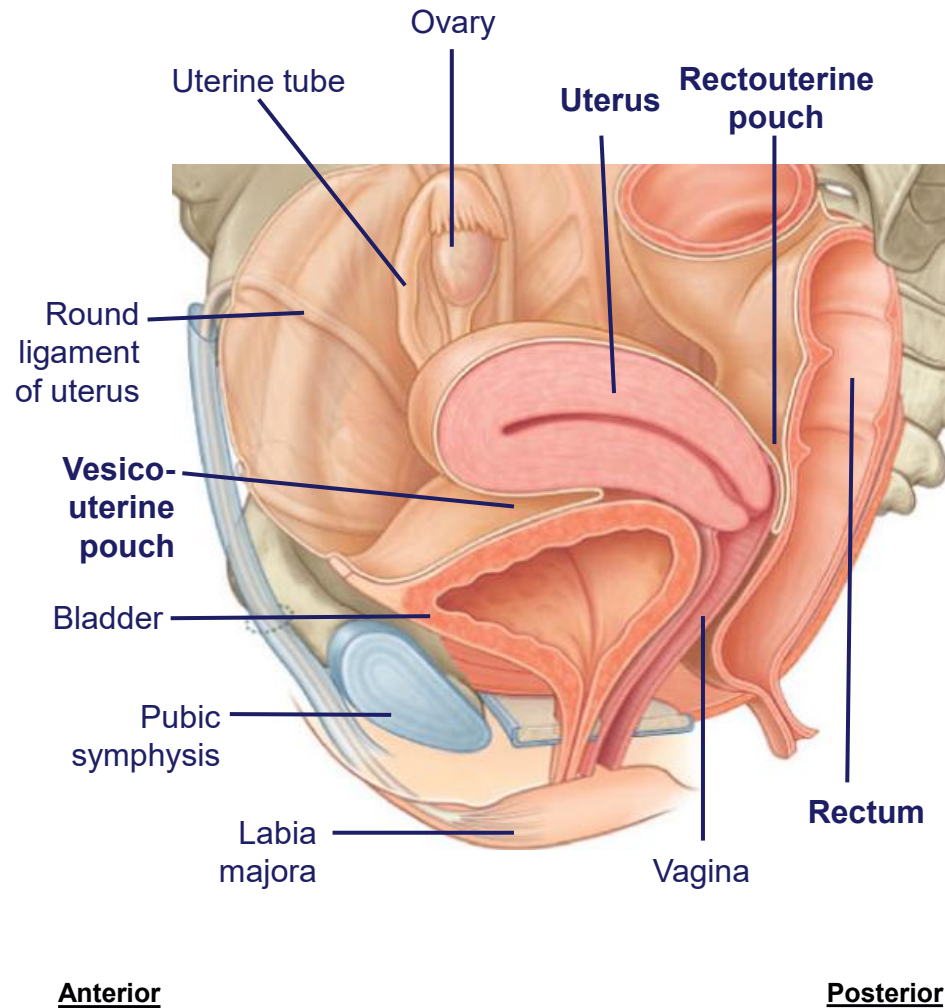
Anatomical relationship of female pelvic organs



Clinical anatomy of the female reproductive organs

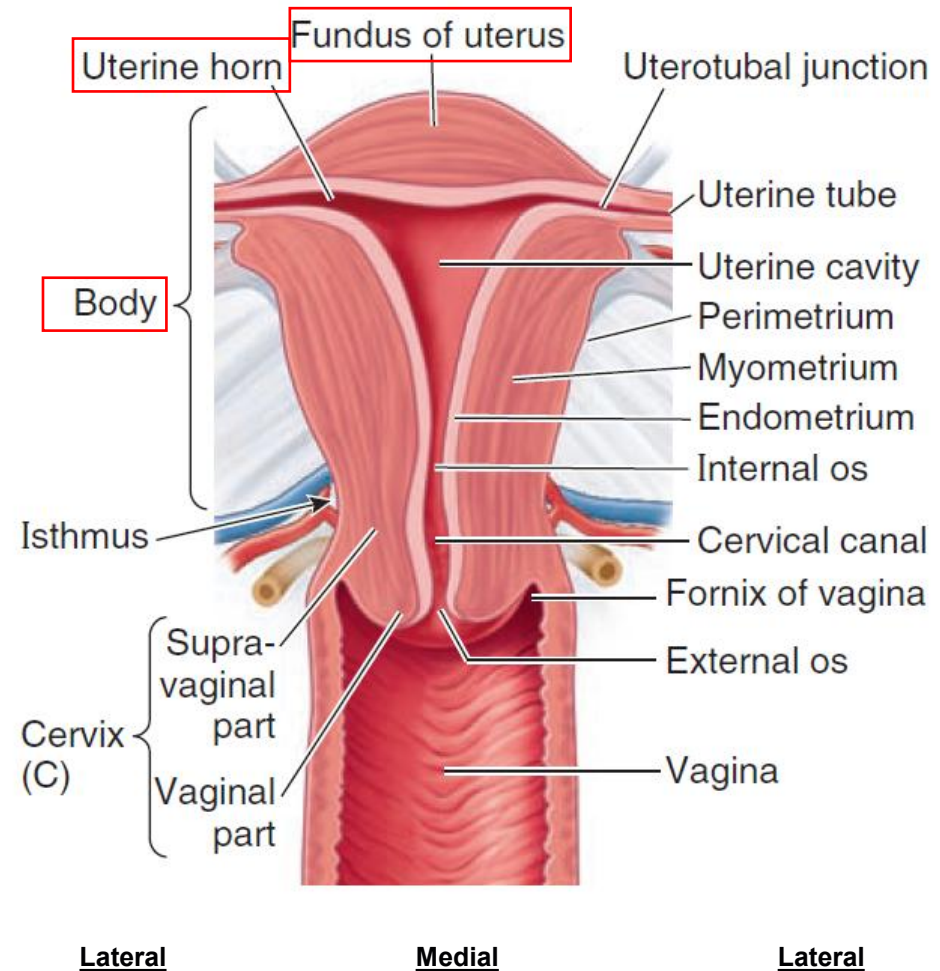
Uterus

- The uterus (womb) is a thick-walled, pear-shaped, hollow muscular organ.
- Its size is about 8 x 5 x 3 cm.
- The **uterus** lies superior and posterior to the bladder.
 - The **vesicouterine pouch**, a peritoneal reflection that separates the posterior aspect of the bladder and the anterior surface of the uterus.
- The **uterine body and cervix** lie **anterior to the rectum**.
 - The **rectouterine pouch (pouch of Douglas)**, a peritoneal reflection that separates the rectum from the uterus and cervix. It is the most dependent part of the peritoneal cavity.



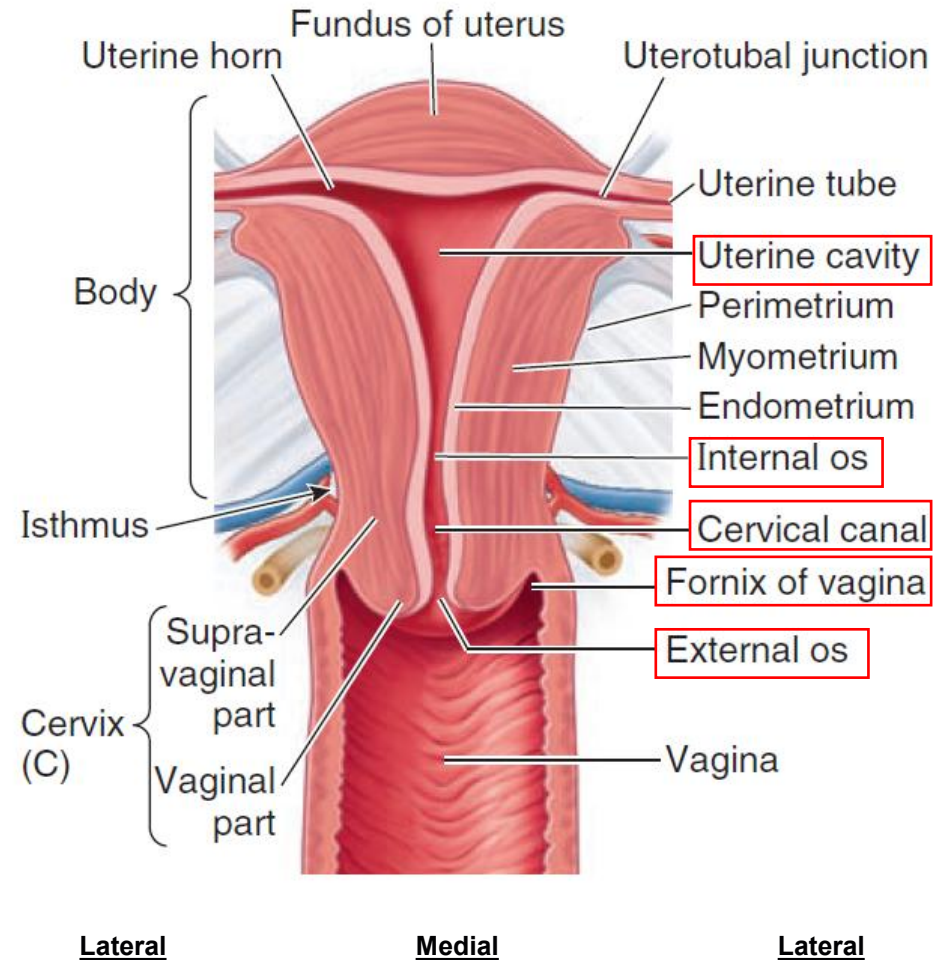
Uterus

- It consists of a **body** and a **cervix**, and inferiorly it joins the vagina
- Body (Corpus)**: The central, largest portion that is flattened anteroposteriorly. It is the site of implantation.
 - Fundus**: rounded, dome-shaped superior end of body above the level of the origin of the uterine tubes.
 - Uterine horns (cornua of uterus)**: the points at which the uterine tubes enter the uterus.
 - Attachment point for:
 - Round ligament of the uterus - anterior to the uterine tube
 - Ovarian ligament - posterior to the uterine tube.



Uterus

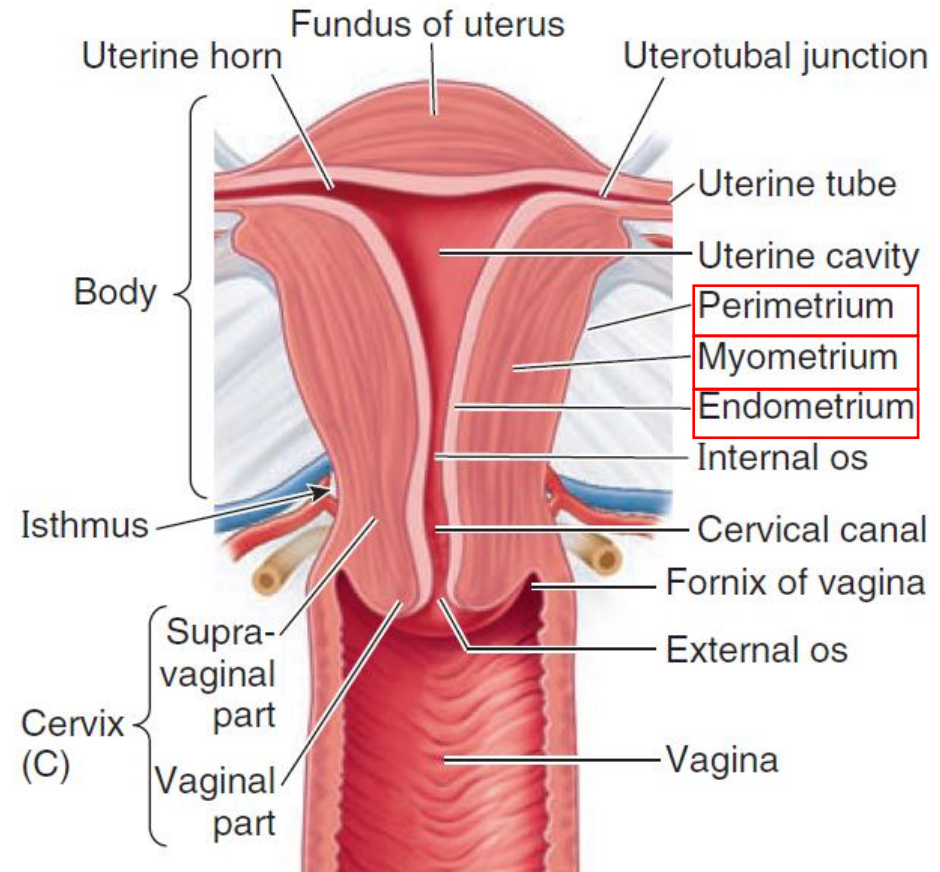
- **Uterine Cavity:** Triangular space with the apex leading to the internal os of the cervix.
 - It is a narrow slit, when viewed laterally, and is shaped like an inverted triangle, when viewed anteriorly.
 - Each of the superior corners of the cavity is continuous with the lumen of a uterine tube.
- **Cervix:** The lower, cylindrical portion that protrudes into the vagina causing the superior most portion of the vagina to dome around it creating a gutter or “**fornix**” in the vagina.
 - **Internal os:** The opening between the uterine cavity and the cervix.
 - **Cervical canal**
 - **External os:** The opening from the cervix to the vagina.



Uterine wall

The uterine wall has three layers:

- **Endometrium:** This is the innermost layer that lines the uterine cavity. It thickens and sheds during the menstrual cycle.
- **Myometrium:** This is the middle layer composed of smooth muscle cells. It contracts during childbirth to help expel the baby.
- **Perimetrium (or Serosa):** This is the outermost layer covering the uterus. It is a continuation of the peritoneum, which is the membrane that lines the abdominal cavity.



Lateral

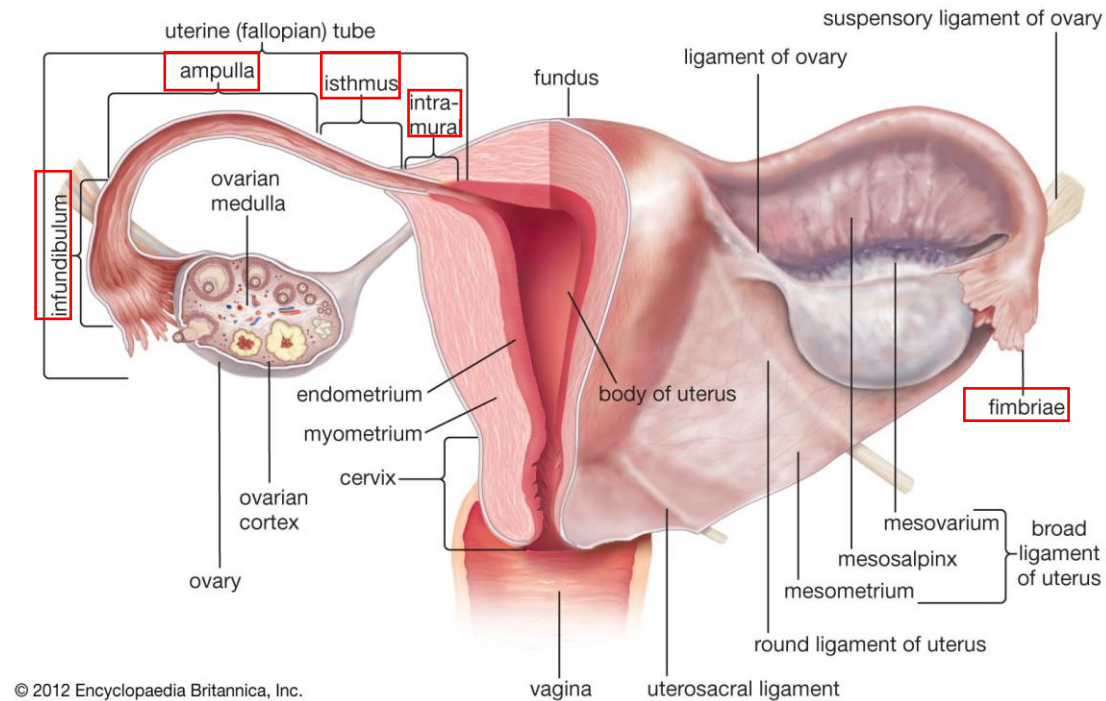
Medial

Lateral

Uterine tubes (oviduct)

Divided into four sections:

- **Interstitial (intrauterine/intramural) part:** This is the section that lies within the wall of the uterus.
- **Isthmus:** This is a narrow section adjacent to the uterus.
- **Ampulla:** This is the longest and widest section of the tube. It's where fertilization typically takes place.
- **Infundibulum:** This is the funnel-shaped portion near the ovary, with finger-like projections called **fimbriae** that help to catch and guide the ovulated egg into the tube.
 - The fimbrial end of the uterine tube is open to the peritoneal cavity.



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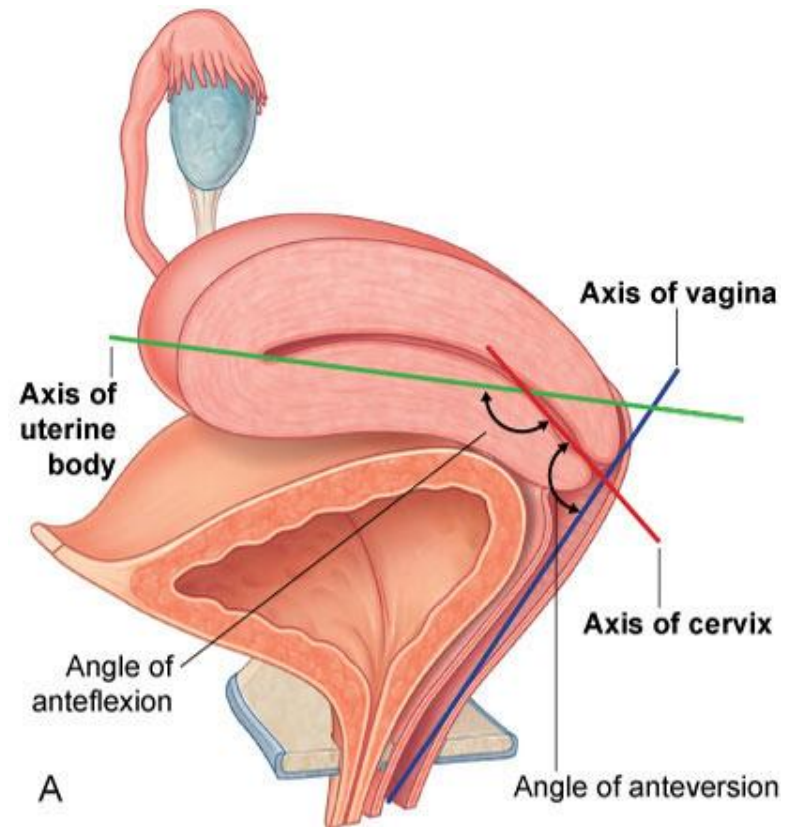
Lateral

Medial

Lateral

Uterine position

- **Uterine version** is defined as **the angle that the cervical axis makes with the vaginal axis:**
 - **Anteversion:** long axis of the cervix forms an acute angle with the long axis of the vagina, such that the uterus tilts forward toward the anterior abdominal wall.
 - **Retroversion:** uterus tilts backward so that its long axis (angle between the cervix and vagina) is directed posteriorly relative to the vagina
- **Uterine flexion** is defined as **the angle that the axis of the uterine body makes with the cervical axis:**
 - **Anteflexion:** uterine body is flexed forward relative to the cervix, so the fundus bends anteriorly over the bladder.
 - **Retroflexion:** body of the uterus is bent backward on itself at the junction of the cervix and the body
- Uterus is usually **anteverted** and **anteflexed**.



Anterior

Posterior

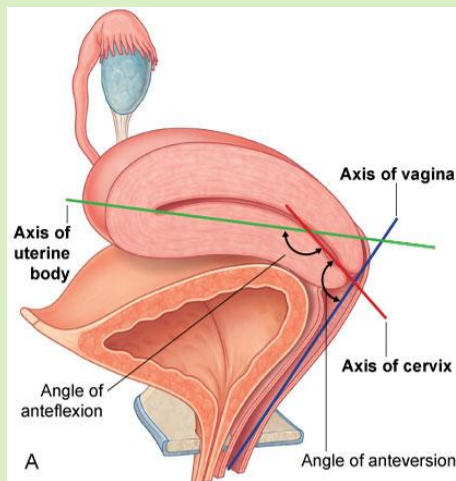
Uterine position

Anteflexed uterus: The anteflexed uterus is the most common position of the uterus, present in about 50% of women. The uterus is bent forward at the junction of the body and the cervix with the uterus curving forward over the bladder.

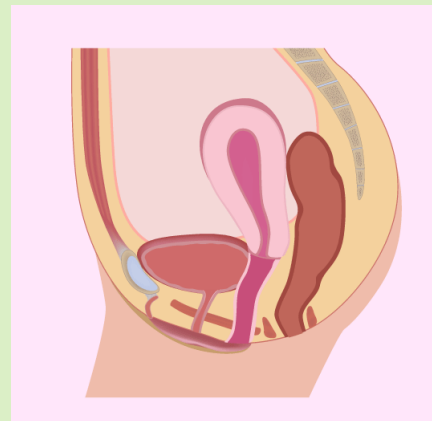
Anteverted uterus: The uterus tilts forward, so the axis of the cervix is angled anteriorly relative to the vagina. Common anatomical variant, usually found with anteflexion.

Retroflexed uterus: A retroflexed uterus is a condition where the body of the uterus is bent backward on itself at the junction of the cervix and the body, while the cervix maintains its usual position with the vagina.

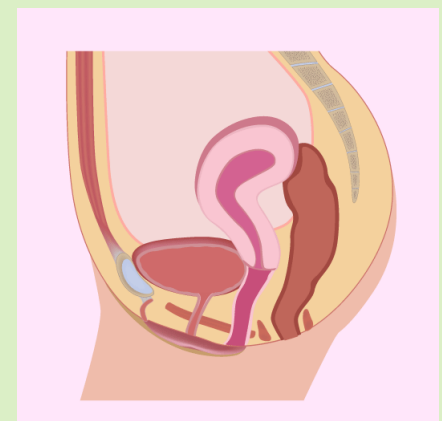
Retroverted uterus: The uterus tilts backward so that its long axis (angle between the cervix and vagina) is directed posteriorly relative to the vagina. The body of the uterus is correctly aligned with the cervix, but the whole structure leans backward toward the rectum.



Anteverted and Anteflexed



Retroverted



Retroverted and Retroflexed

Round Ligament

- A fibromuscular band, remnant of gubernaculum
- Helps **maintain the anteflexed and anteverted position of the uterus** (tilted forward).

Origin:

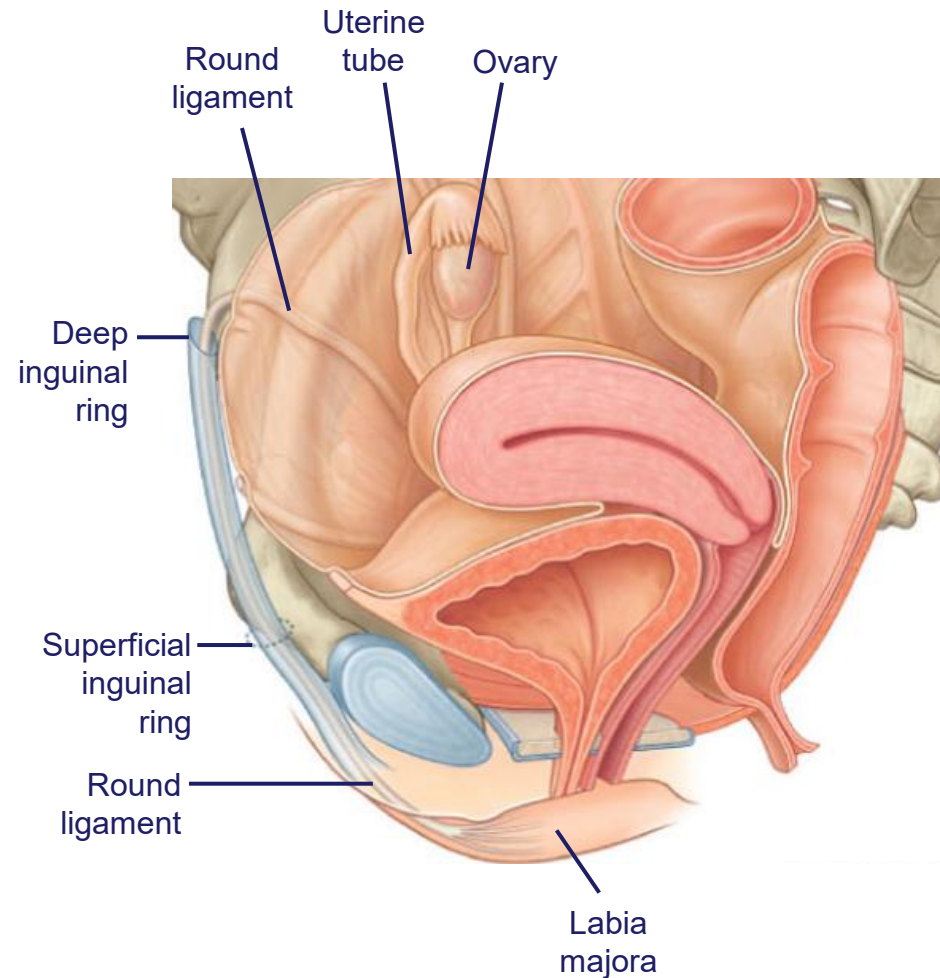
- Arises from the uterine horns, near the junction of the uterus and uterine tubes.

Course:

- Passes anteriorly through the broad ligament to enter the inguinal canal through the deep inguinal ring.
- Exits the inguinal canal through the superficial inguinal ring and terminates in the labia majora, where it blends with connective tissue.

Structure:

- The round ligament consists of smooth muscle fibers, connective tissue, and some blood vessels and nerves.

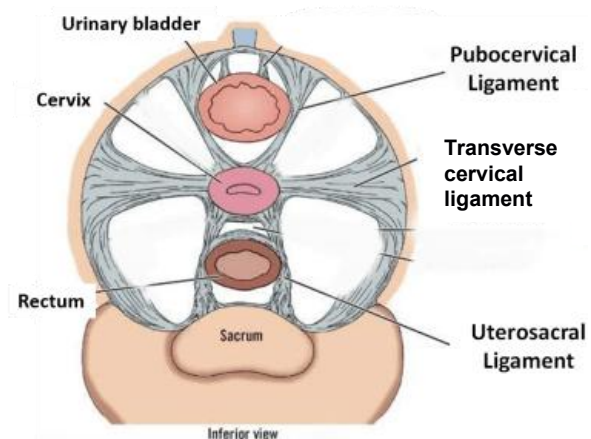
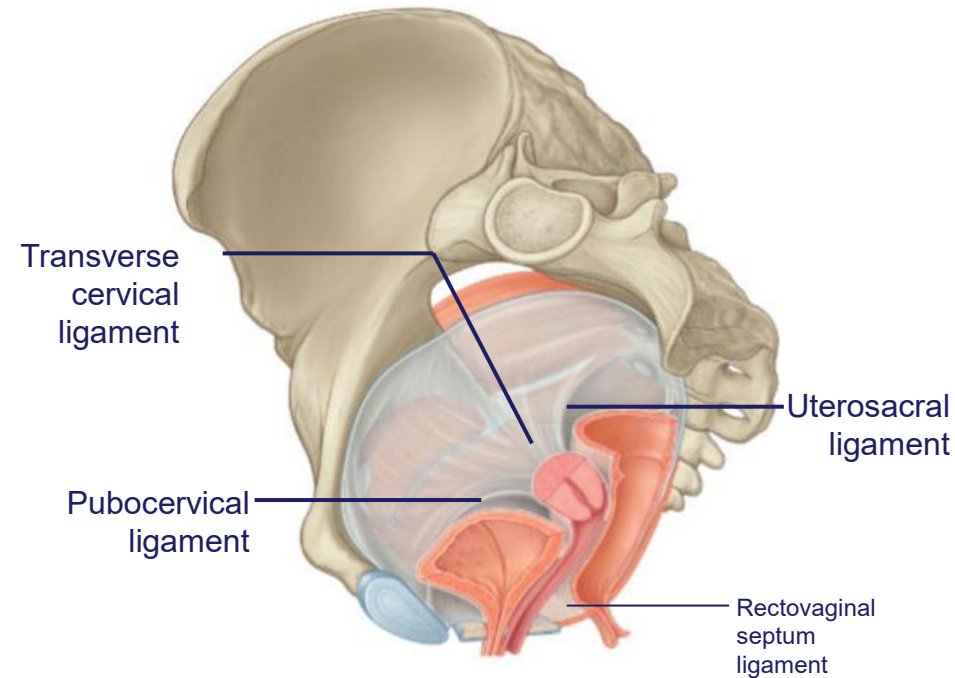


Anterior

Posterior

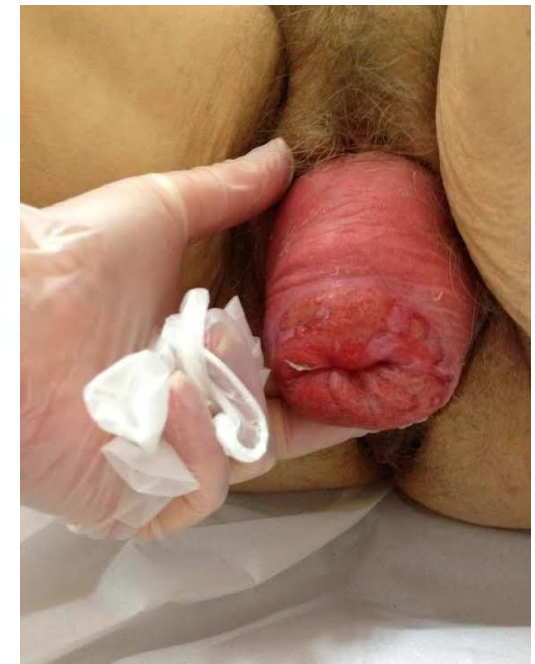
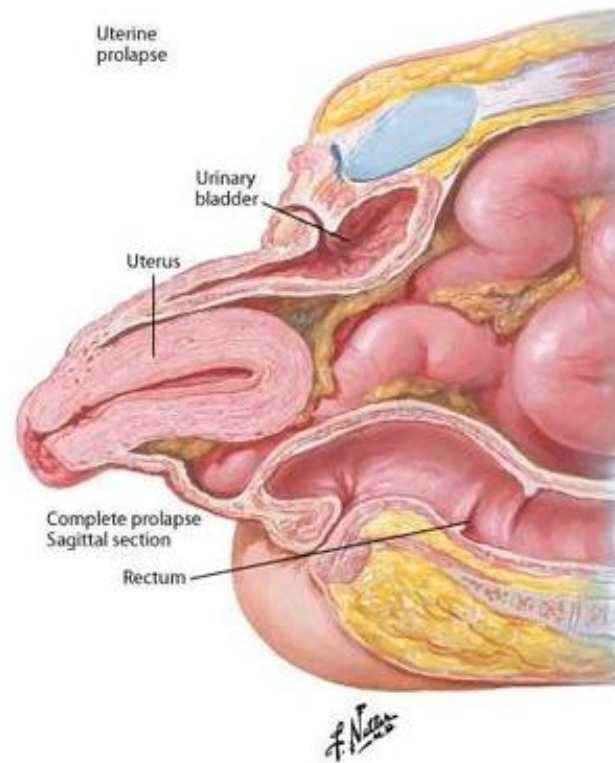
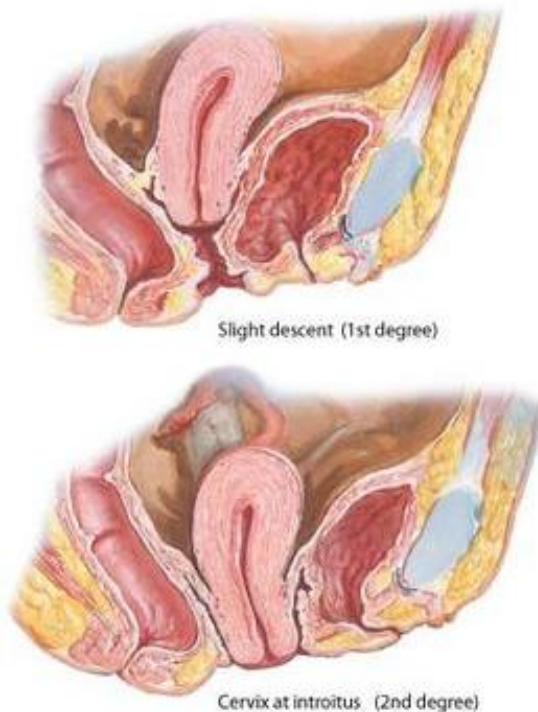
Supportive ligaments of the uterus and cervix

- Condensations of pelvic fascia forms ligaments that extend from the uterus and cervix to the pelvic walls.
- They function to **stabilize the uterus and cervix in the pelvic cavity.**
- **Pubocervical ligament**
 - Extends from the anterior aspect of the cervix and upper vagina to the inner aspect of the pubic bone.
- **Transverse cervical (Cardinal) ligament**
 - Extends from supravaginal part of cervix and upper vagina to the lateral pelvic wall. Found at the base of the broad ligament. Carries the uterine artery.
- **Uterosacral ligament**
 - Extends from the posterior sides of the cervix and uterine body to the anterior aspect of sacrum



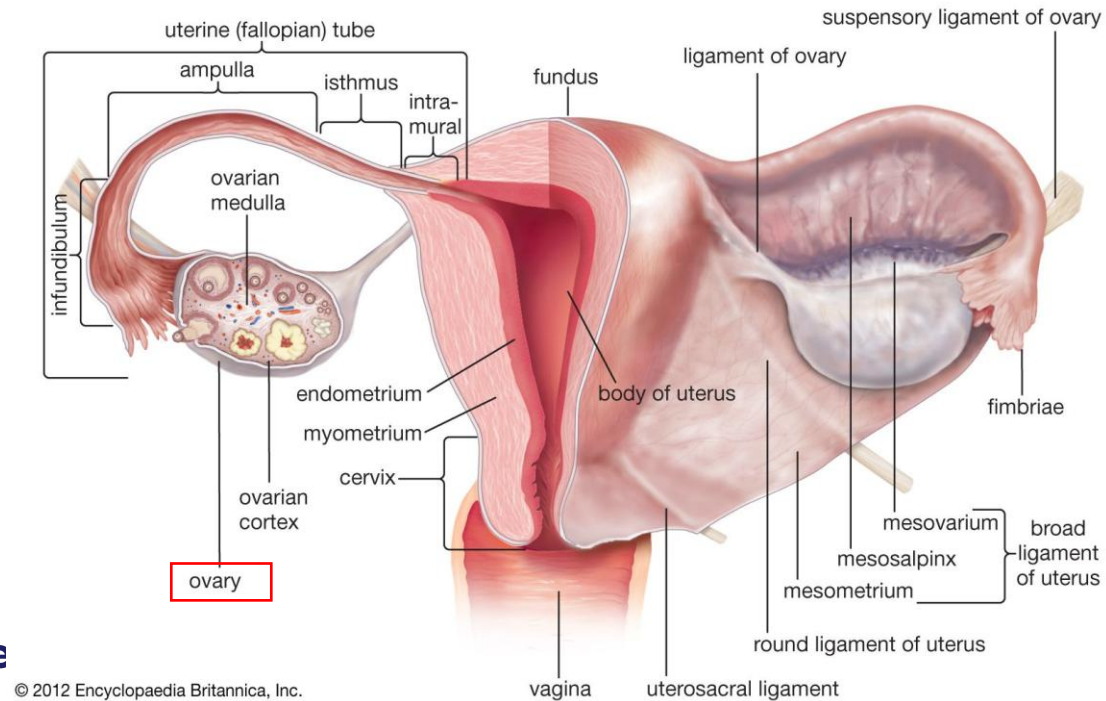
Clinical correlate: uterine prolapse

- Due to the loss of support by the cardinal/uterosacral/pubocervical ligaments and/or by levator ani.
- Support structures can no longer counteract typical intra-abdominal pressures.
- Various degrees of prolapse may occur.



Ovaries

- A pair of small, almond-shaped organs located in the pelvic cavity of females, on either side of the uterus in the ovarian fossa.
- In an adult woman, each ovary is roughly 3-5 cm in length, 1.5-3 cm in width, and 0.6-1.5 cm in thick.
- They developed in the abdominal cavity and **descend** into the pelvis
- **Takes the blood supply, venous drainage, lymphatic drainage and innervation as they descend (located in the suspensory ligament)**
- Produces ova and releases hormones.



Ovaries

Ligamentous Attachments:

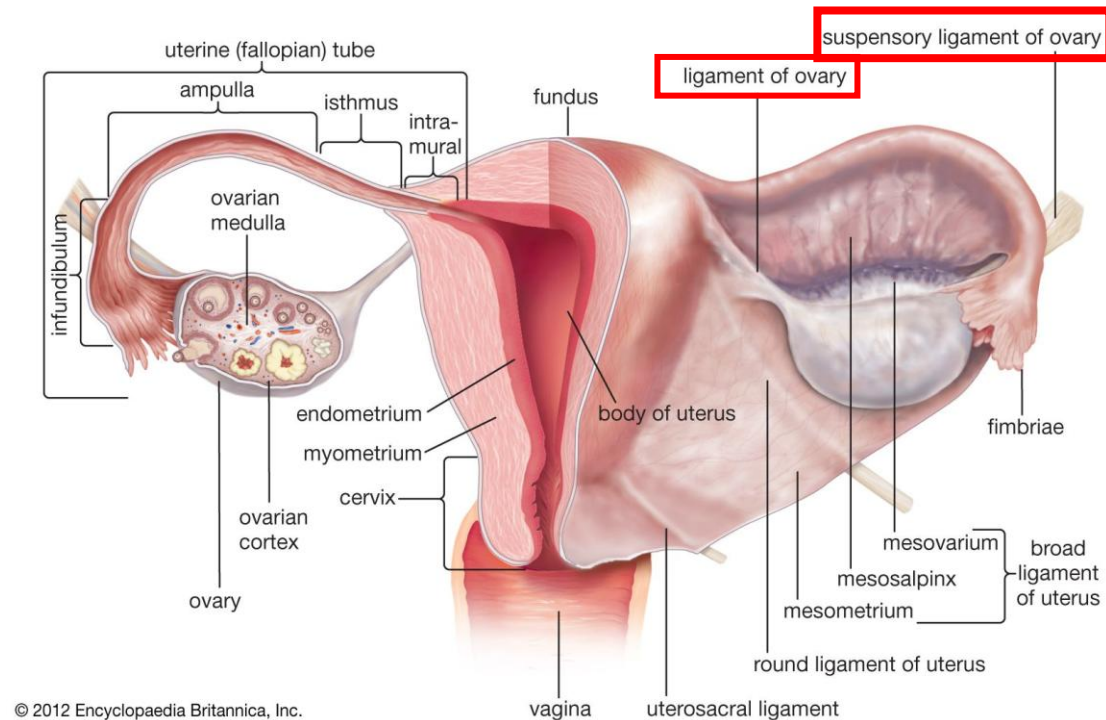
The ovary is connected to surrounding structures by two ligaments:

- **Suspensory ligament of the ovary (infundibulopelvic ligament):**

- This is a raised fold of peritoneum extending from the ovary to the lateral wall of the pelvis. **It contains the ovarian artery, veins, autonomic nerves and lymphatics.**

- **Ovarian ligament proper:**

- This is a fibrous ligament that extends from the cornu of the uterus to the ovary, anchoring it to the uterus.



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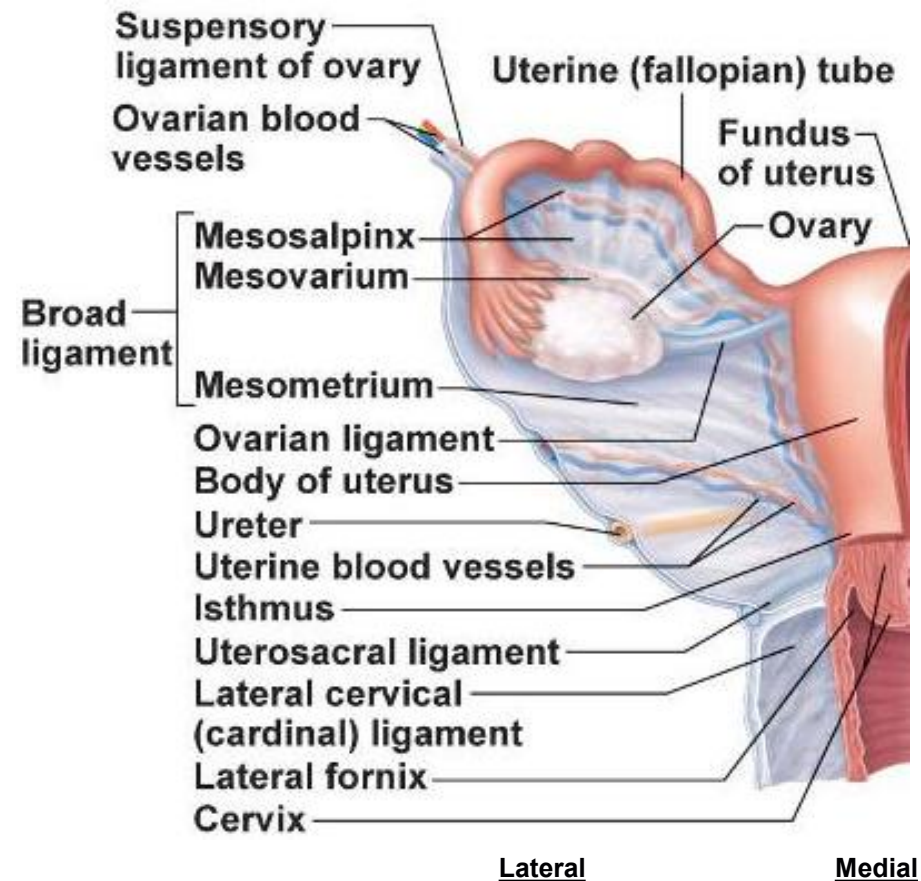
Peritoneal folds - broad ligament

A sheet-like fold of peritoneum, extending from the sides of uterus to the lateral pelvic wall.

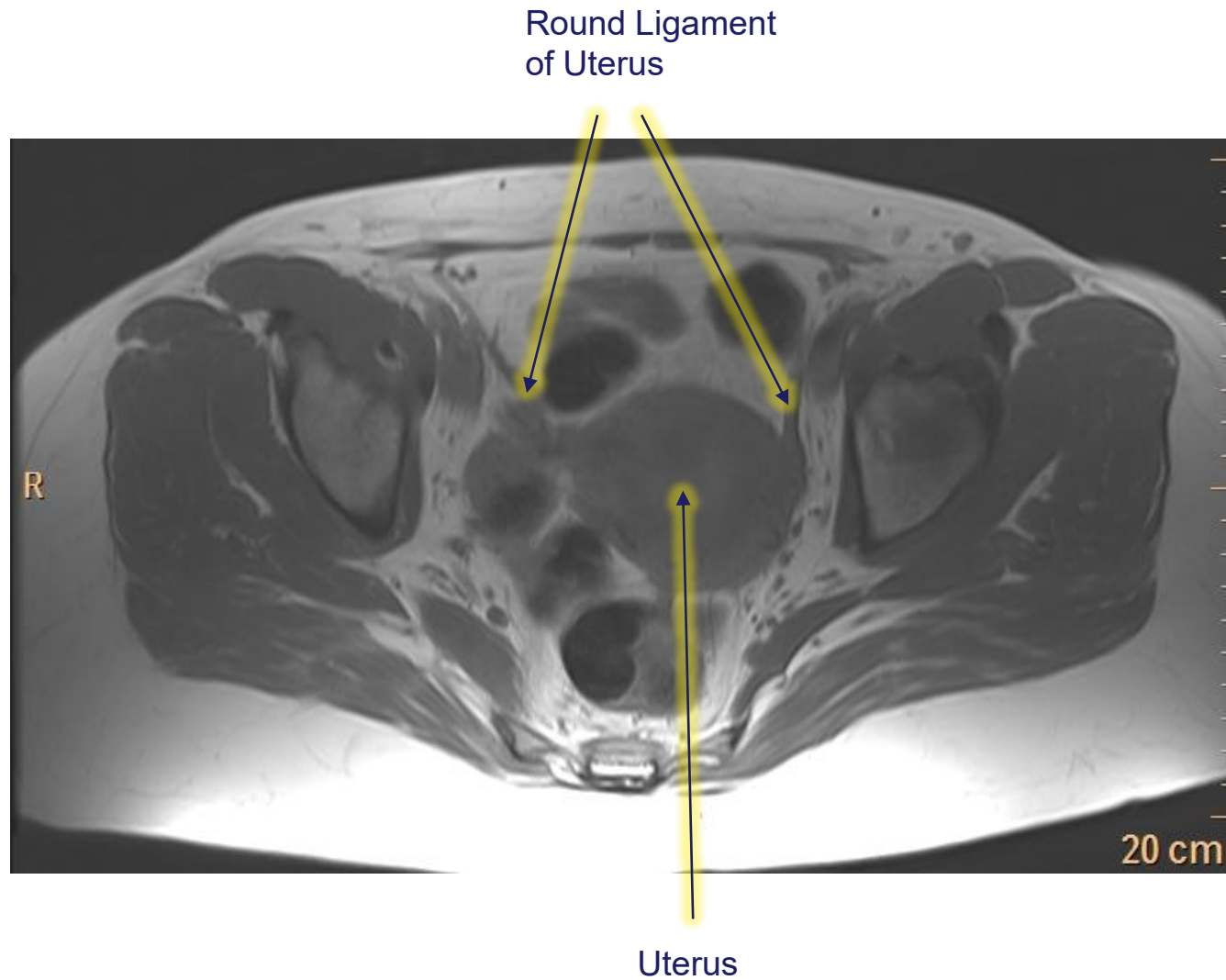
Encloses the uterus (except the cervix), the uterine tubes in its superior margin and suspends the **ovary** from its posterior aspect.

The broad ligament has three parts:

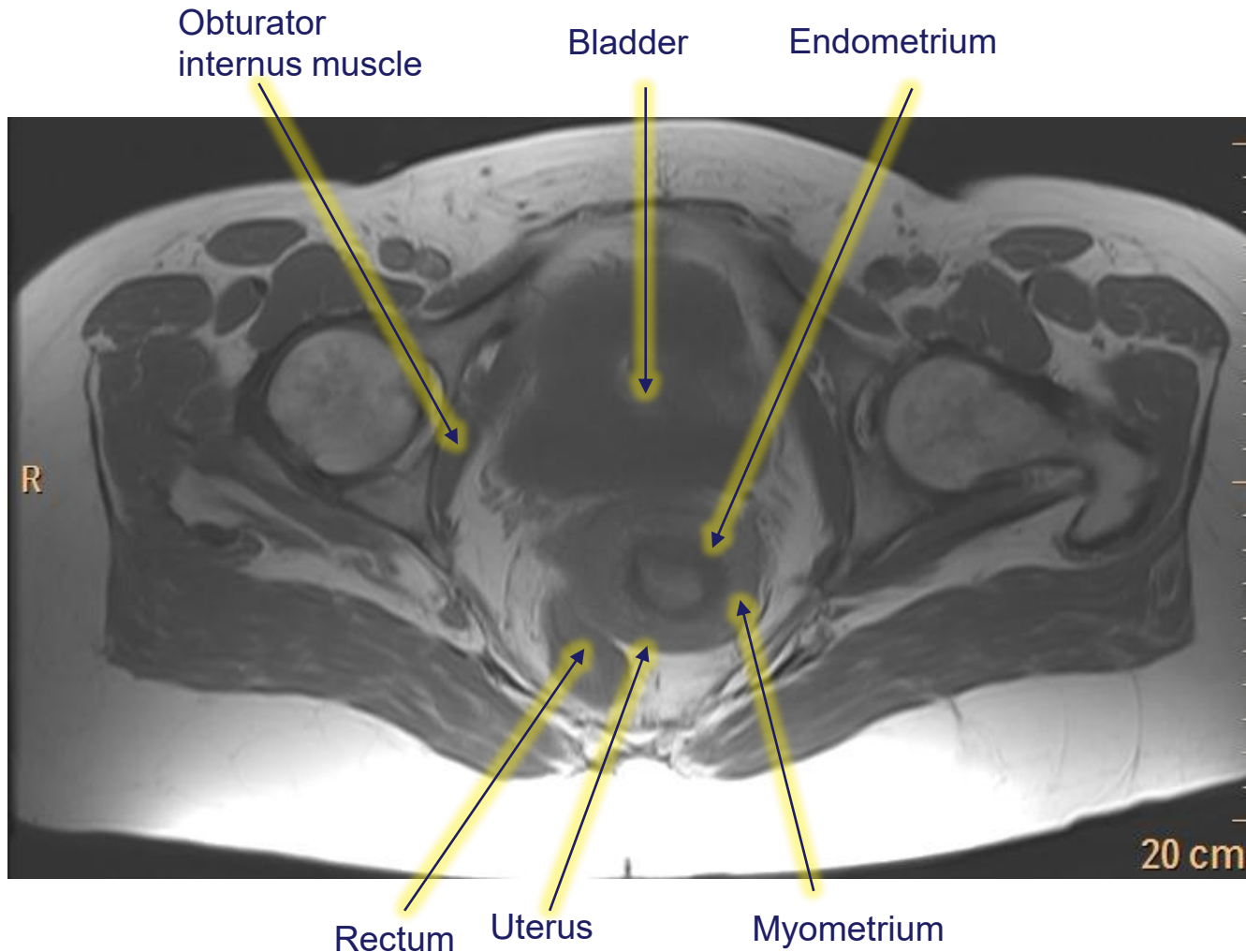
- The **mesometrium**, the largest part of the broad ligament, which extends from the lateral pelvic walls to the body of the uterus
- The **mesosalpinx**, the most superior part of the broad ligament, which suspends the uterine tube in the pelvic cavity
- The **mesovarium**, a posterior extension of the broad ligament, which attaches to the ovary. The peritoneum of the mesovarium is continuous with the ovarian surface (germinal) epithelium.



MRI of female reproductive organs

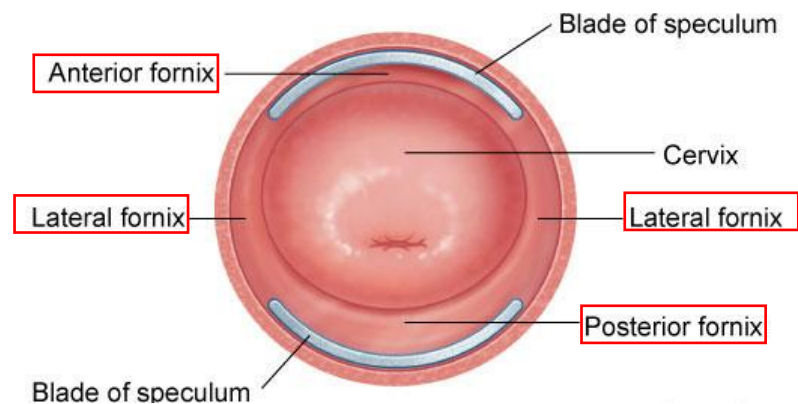
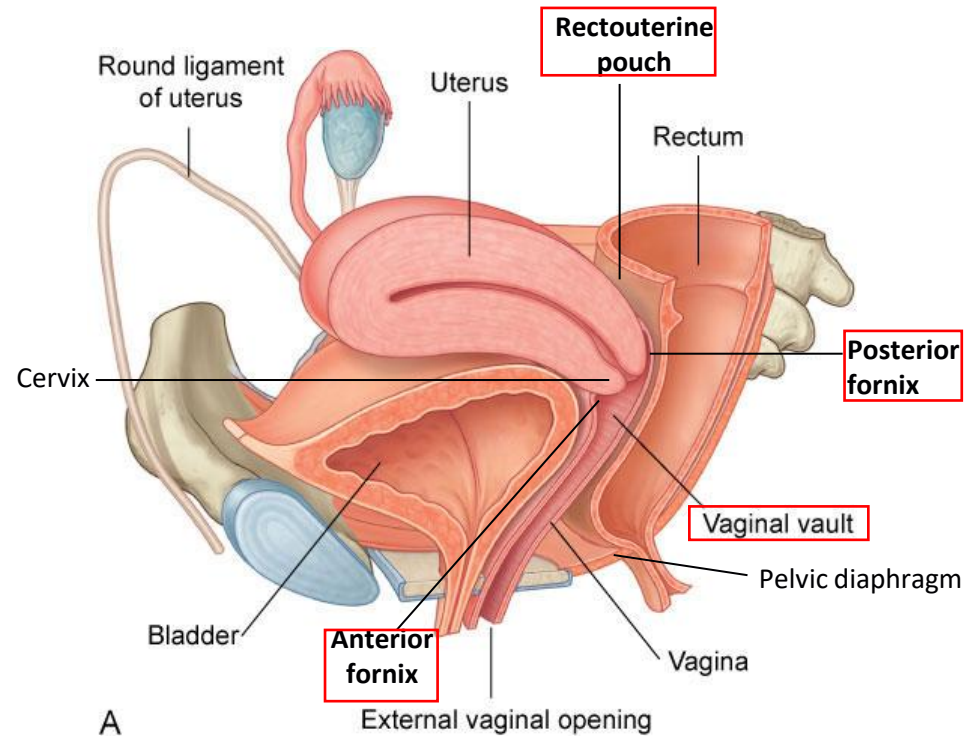


MRI of female reproductive organs



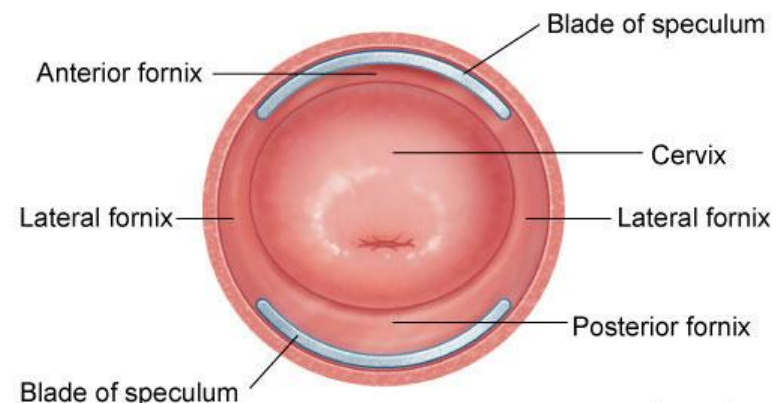
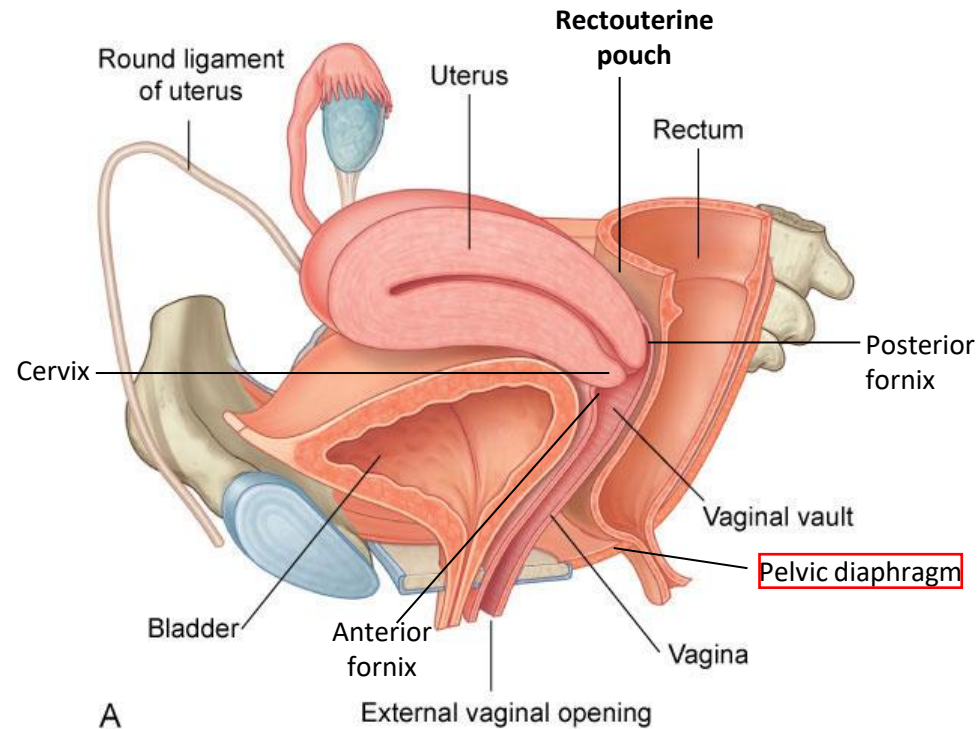
Vagina

- It is a distensible fibromuscular tube that extends from its external opening (the **introitus**) in the perineum into the pelvic cavity attaching to the cervix.
- The internal end of the canal is enlarged to form a region called the **vaginal vault**.
- The **vaginal fornix** is the recess formed between the margin of the cervix and the vaginal wall.
 - Based on position, the fornix is subdivided into a **posterior** fornix, an **anterior** fornix, and two **lateral** fornices.
 - The **posterior fornix** is closely associated with the **rectouterine** pouch

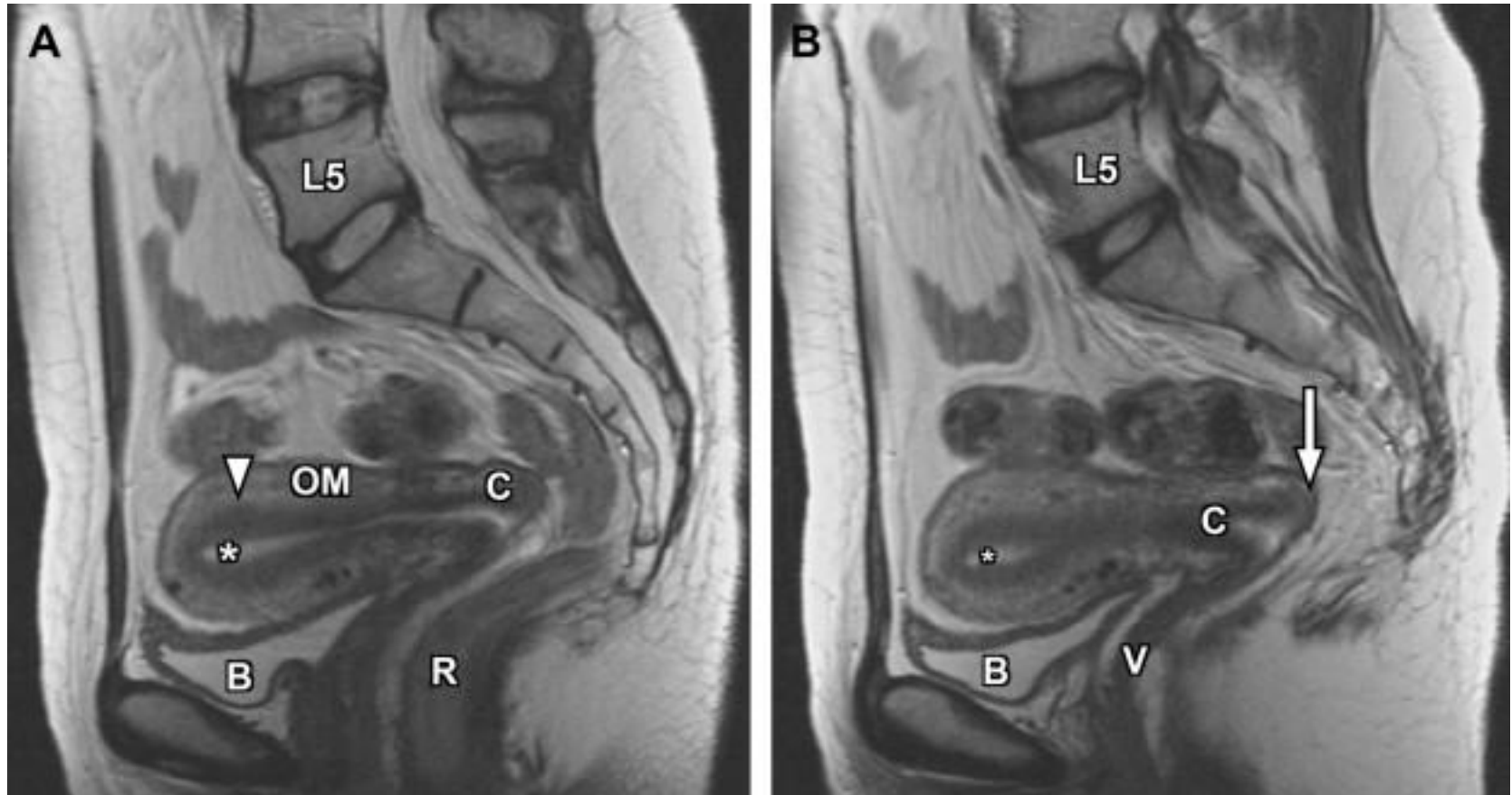


Vagina

- **Pelvic part:**
 - **upper two thirds** lying above the pelvic diaphragm
- **Perineal part:**
 - **lower one third** lying beneath the pelvic diaphragm
- The **anterior wall** of the **vagina** is **posterior** to the **base of the bladder** and to the **urethra**.
- The **posterior wall** of the **vagina** is **anterior** to the **rectum**.
 - **Rectovaginal fascia (septum)** layer of connective tissue separating the rectum from the vagina (**important for preventing rectocele**).
- **Inferiorly**, the **vagina opens** into the **vestibule** of the perineum immediately **posterior** to the external opening of the **urethra**.



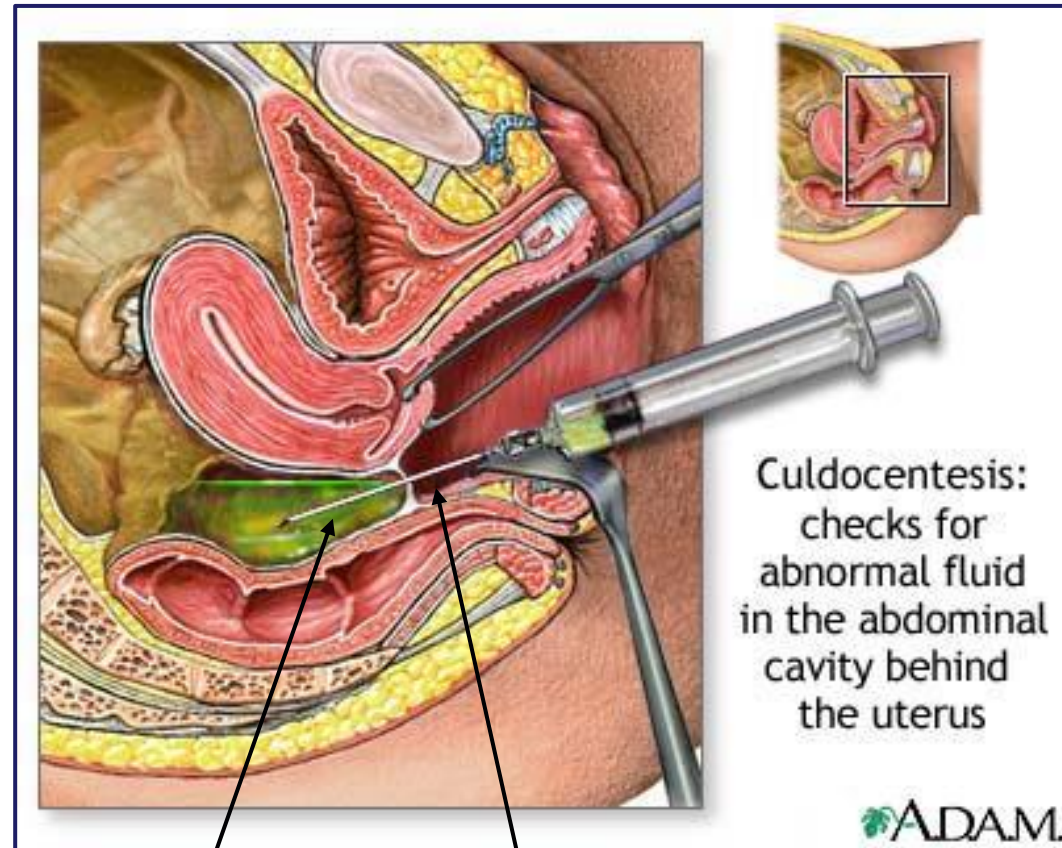
MRI of female reproductive organs



B, bladder; C, cervix; L5, L5 vertebral level; OM, outer myometrium; R, rectum; V, vagina.

Clinical correlate: culdoscopy & culdocentesis

- **Culdoscopy:** minimally invasive endoscopic procedure where a culdoscope is inserted through the posterior vaginal fornix to examine the rectouterine pouch, ovaries and uterine tubes. This procedure can be used for diagnosing pelvic diseases and assessing fertility.
- **Culdocentesis:** medical procedure where fluid is aspirated from the rectouterine pouch through the posterior vaginal fornix using a needle. It's primarily used to detect blood in the abdominal cavity particularly in suspected cases of ruptured ectopic pregnancy.

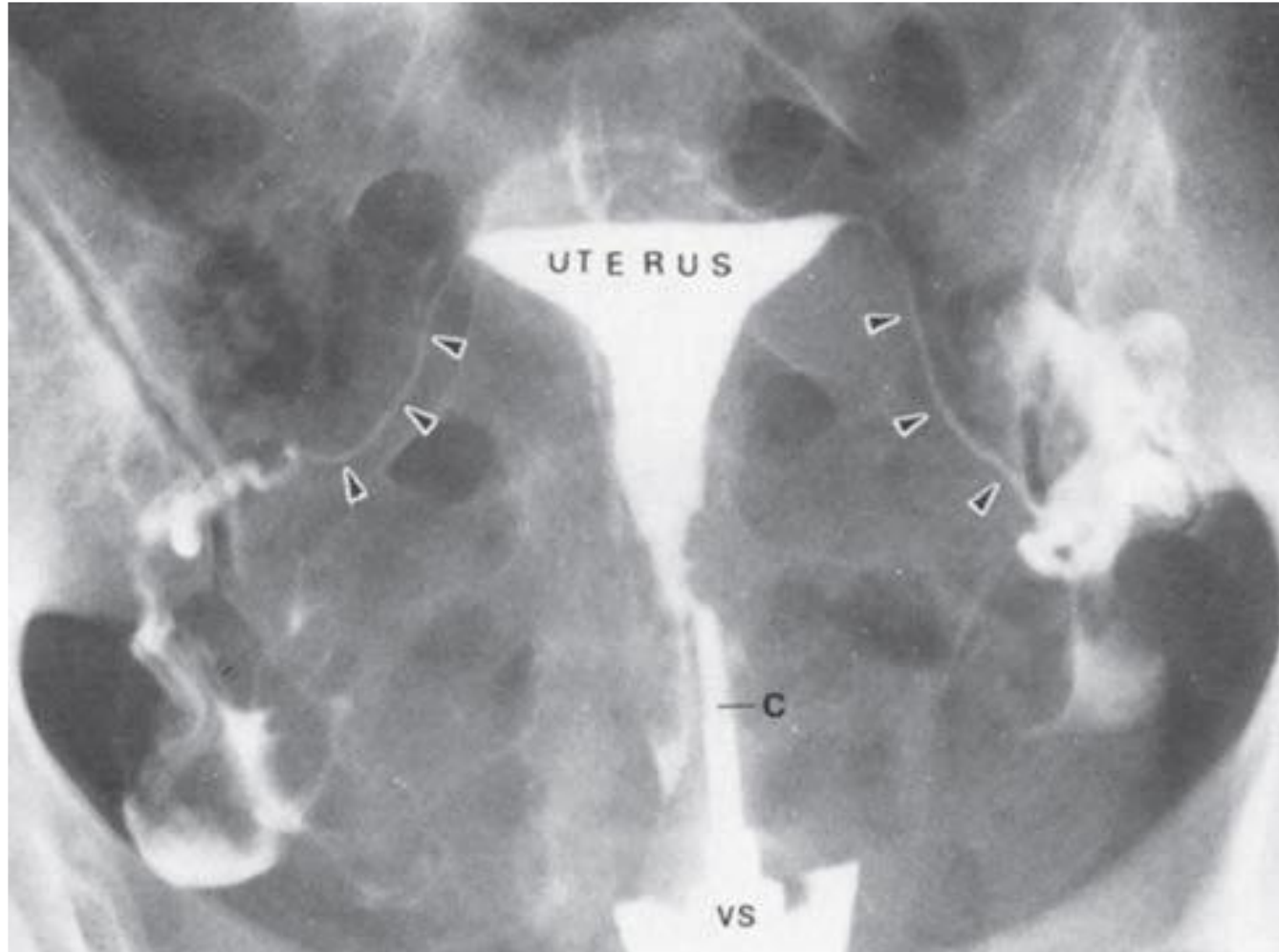


Rectouterine
pouch

Posterior
fornix

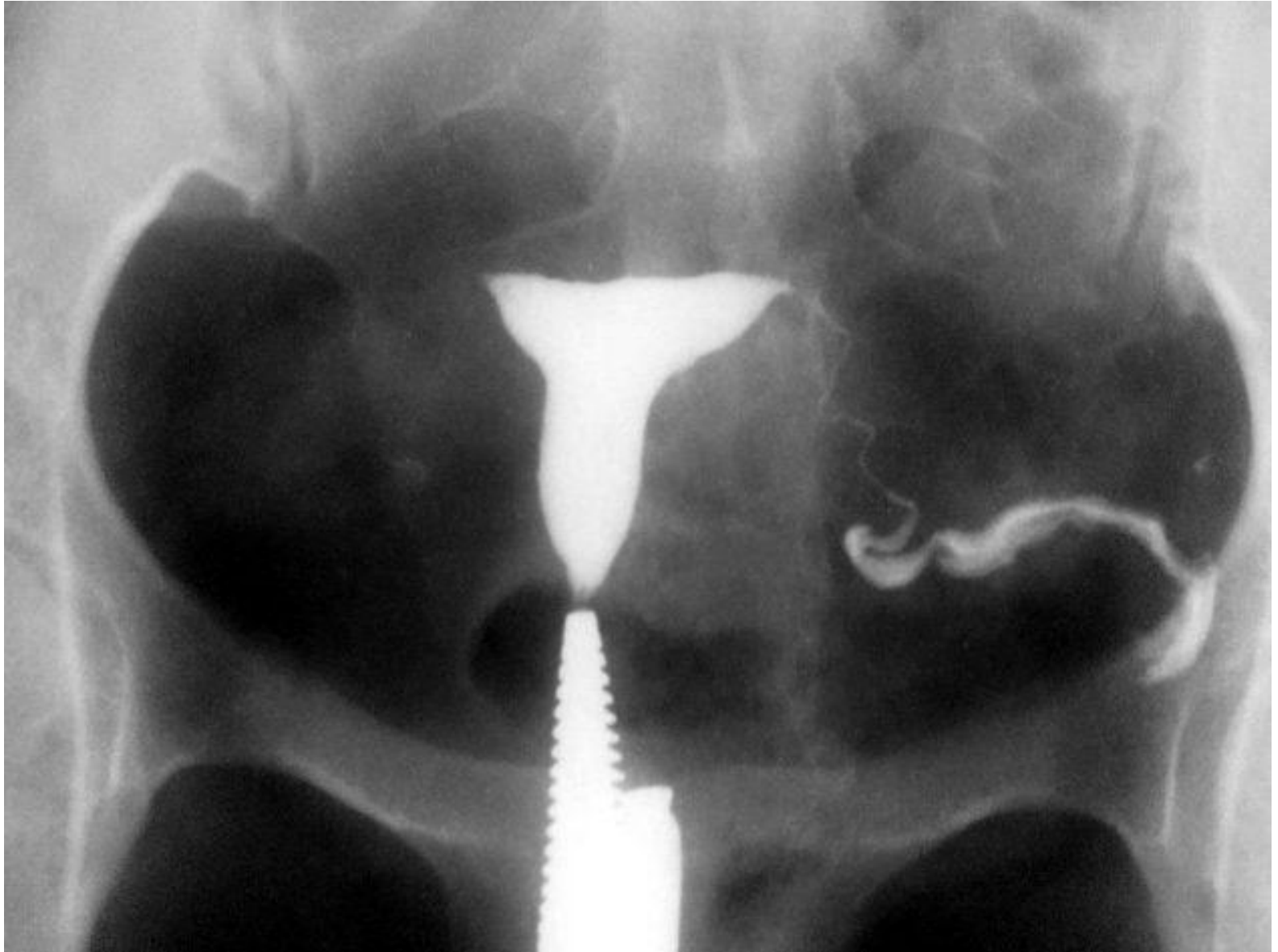
Hysterosalpingogram

- Contrast material injected into the uterus via the cervix



Arrowheads, uterine tubes; *c*, catheter in the cervical canal; *vs*, vaginal speculum

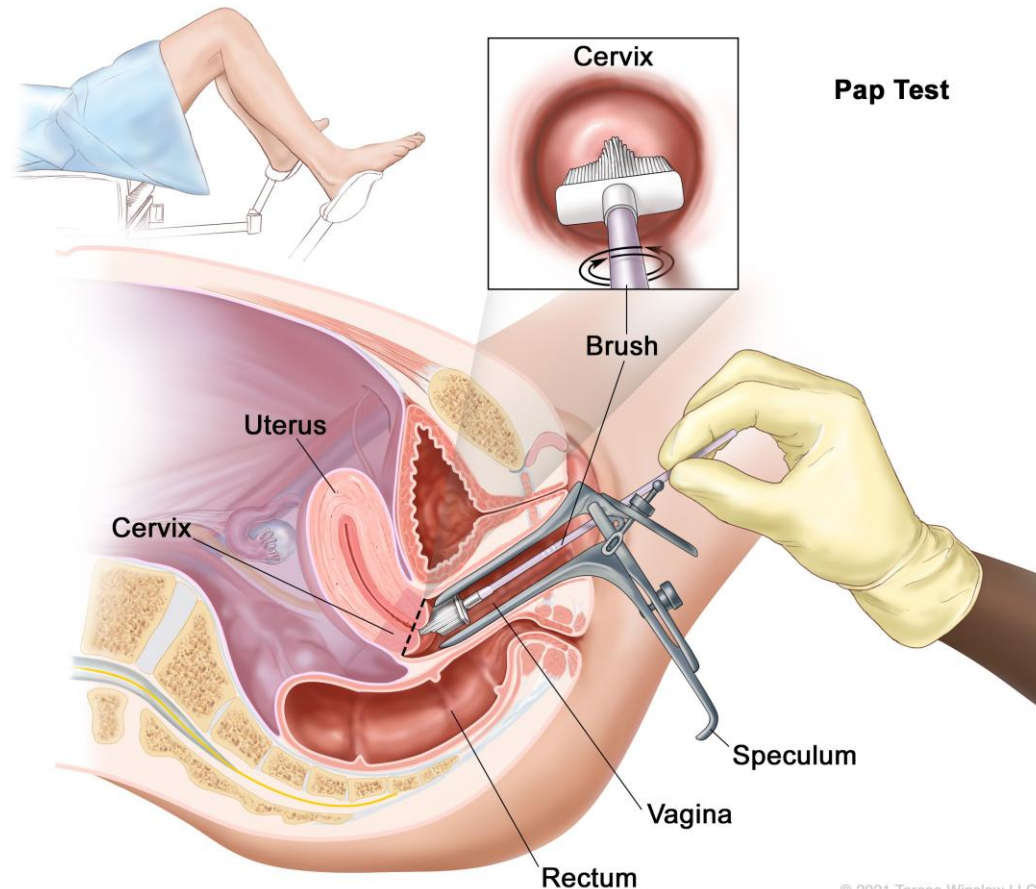
Can you identify the abnormality?



Case courtesy of Dr Kewal Arunkumar Mistry, Radiopaedia.org, rID: 35635

Clinical correlate: Papanicolaou (Pap) test

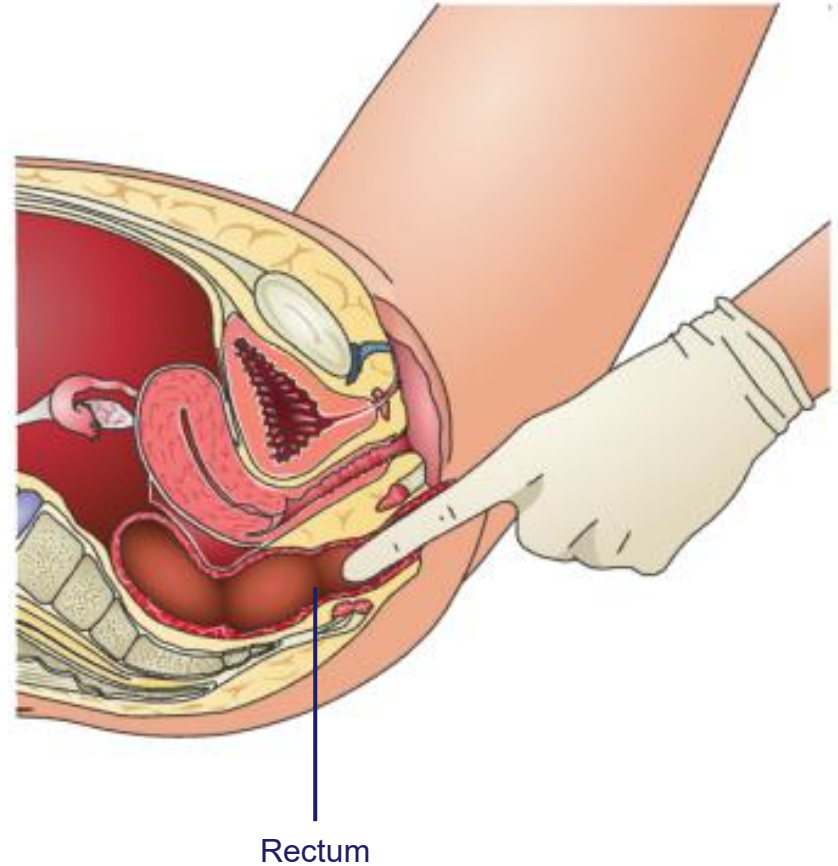
- The Papanicolaou (**Pap**) test is usually done in conjunction with a pelvic exam. In women older than age 30, the Pap test may be combined with a test for human papillomavirus (HPV) — a common sexually transmitted infection that can cause cervical cancer.
- Doctors generally recommend repeating Pap testing every three years for women ages 21 to 65.



Clinical correlate: digital rectal exam (DRE)

- Digital rectal exam can be used to evaluate the female pelvic organs and the pelvic floor.
- Structures that can be palpated in the female are posterior vaginal wall, posterior surface of the uterus and enlarged ovaries.

http://www.patedu.com/english/interactive/pelvic-support-problems-in-women/section6_page5

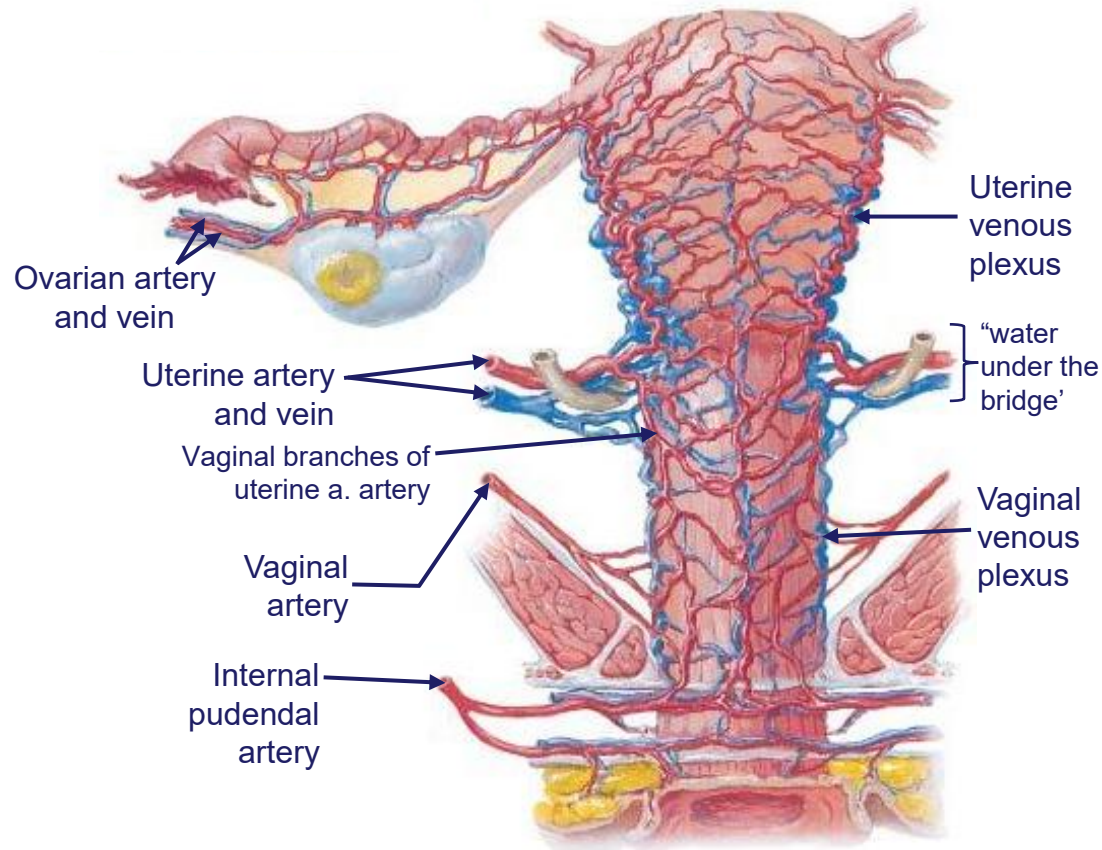


Blood supply and venous drainage of the female reproductive organs

Ovaries, Uterus, Vagina

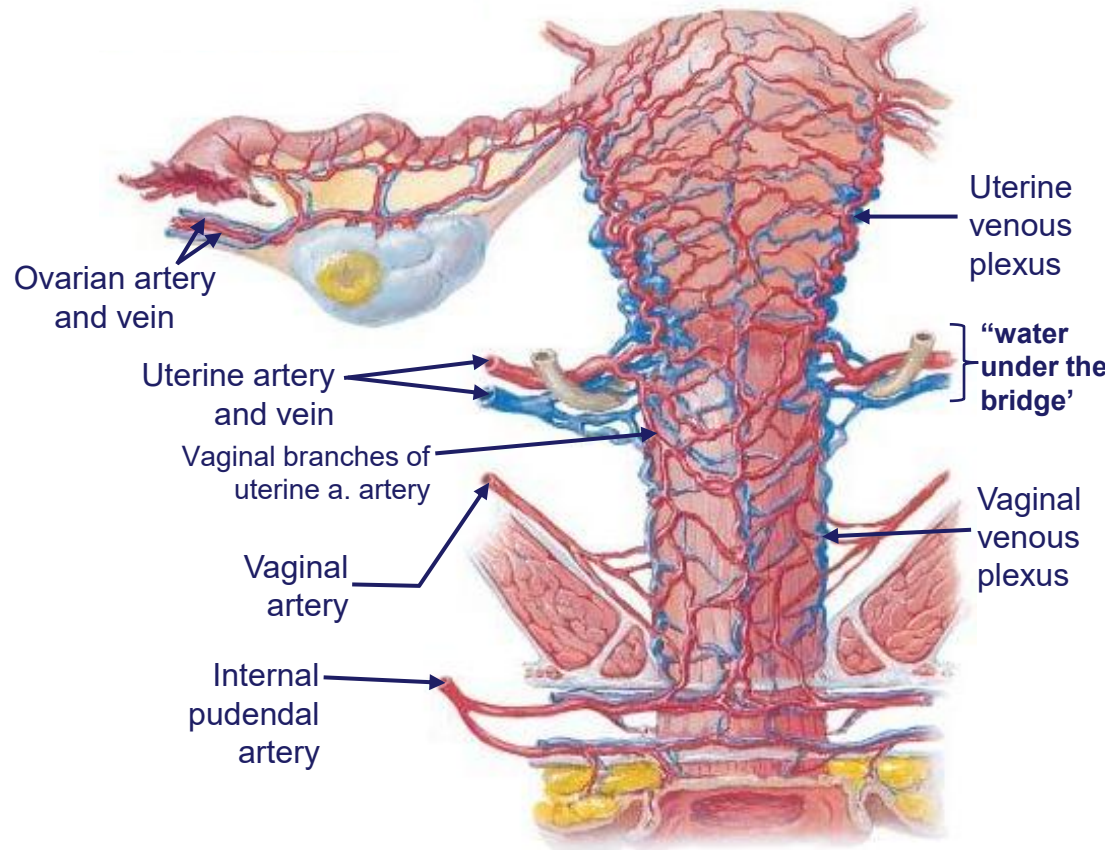
Arterial supply

- **Ovarian artery (branch of abdominal aorta, located in suspensory ligament):** Main blood supply to the ovaries, uterine tubes and fundus. It forms an anastomosis with the uterine artery.
- **Uterine artery (branch of internal iliac):** Main blood supply to the uterus. It also gives off vaginal branches that supply the superior part of the vagina. It forms an anastomosis with the ovarian artery.
- **Vaginal artery (branch of internal iliac):** This is the main source of blood for the vagina.
- **Internal pudendal artery (branch of internal iliac):** This artery provides a portion of the blood supply to the perineal part of the vagina.



Clinical correlate: “water under the bridge”

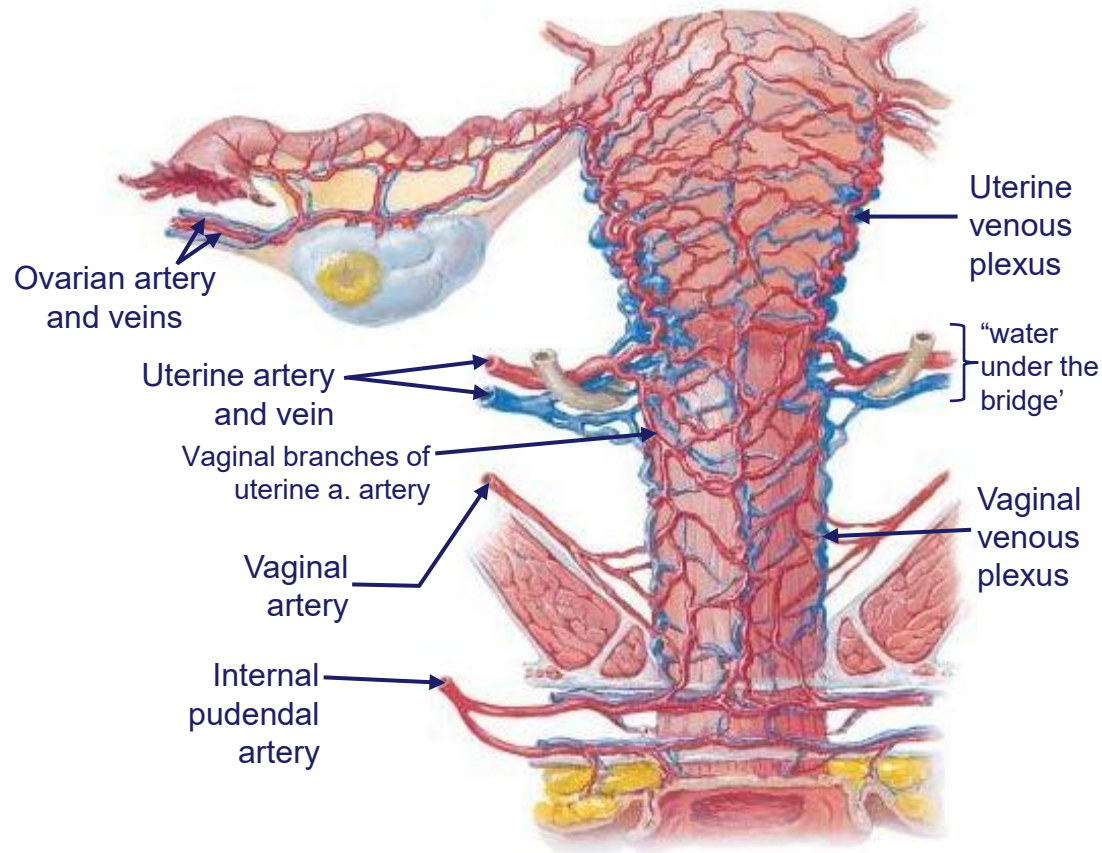
- The uterine artery crosses above the ureters anteriorly in the base of the broad ligament within the **Cardinal ligament**.
- This anatomical relationship is known as **“water under the bridge”** (urine in the ureters is the “water” and the uterine artery is the “bridge”).
- It is important for surgeons performing **pelvic surgeries** such as **hysterectomies**, as they need to be careful **not to damage the ureter** when working near the **uterine artery**.



Ovaries, Uterus, Vagina

Venous drainage

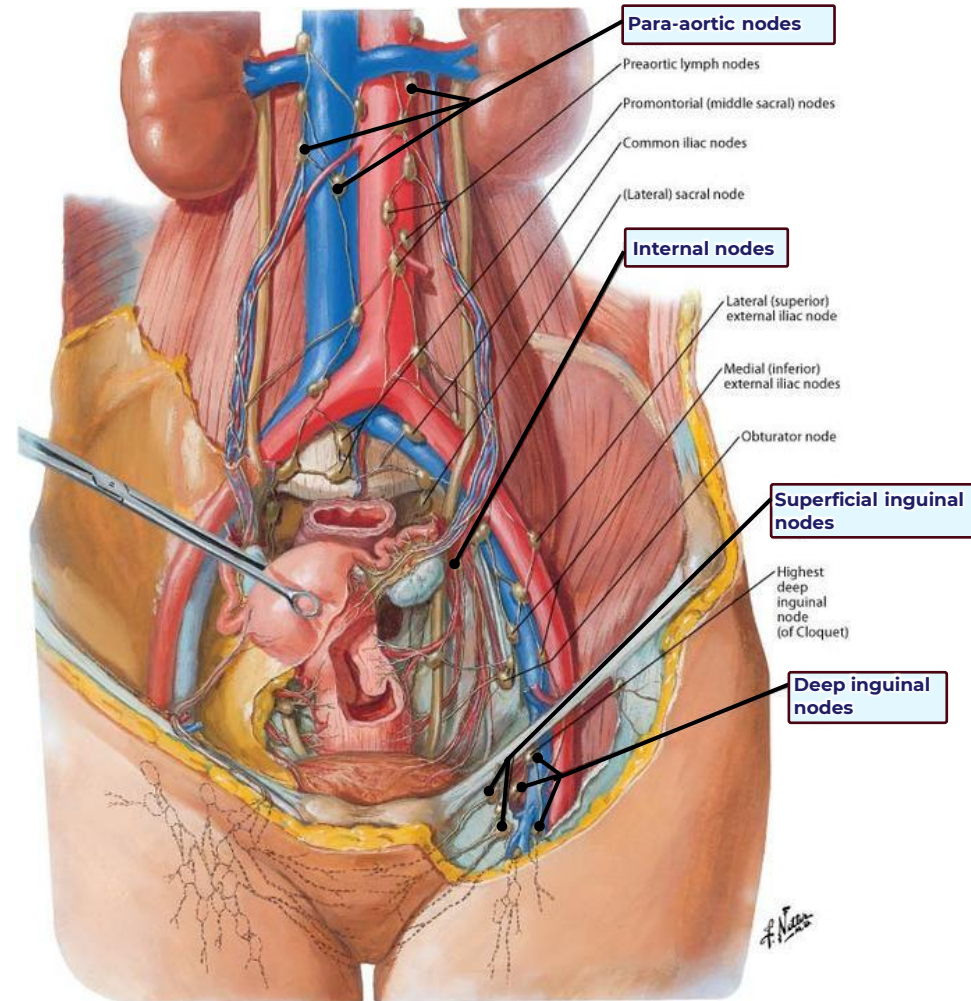
- Within the pelvic cavity, extensive interconnected venous plexuses are associated with the surfaces of the viscera (**uterus, ovary, vagina**).
- They terminate as named veins that correspond to the arteries (uterine vein, ovarian vein, vaginal vein)
- The **uterine and vaginal** veins return to the **internal iliac vein**.
- The **ovarian vein** travels through the **suspensory ligament** back to the posterior abdominal wall to empty into the **inferior vena cava on the right and the renal vein on the left**.



General lymphatic drainage of the female reproductive organs

General lymphatic drainage

- **Ovaries, uterine tubes** and **uterine fundus** of the uterus drain to the **para-aortic (mainly lumbar)** nodes
- **Uterine body** and **cervix** drains to mainly to the **internal iliac** nodes.
- The area of the **uterine cornu** (attachment site for the round ligament) can drain along the **round ligament** to the **superficial inguinal** nodes due to insertion of the round ligament on labia majora.
- The **upper two thirds of the vagina** drains primarily to the **internal iliac** nodes.
- The **lower third of the vagina** is in the perineum and therefore drains to the **superficial inguinal** nodes
 - the **opening into the vestibule** drains to the **deep inguinal** nodes.



Work sheet – Self review on lymphatic drainage of the pelvis and perineum

Nodes	Male Structures	Female Structures
Para-aortic lymph nodes		
Internal iliac nodes		
Superficial inguinal nodes		
Deep inguinal nodes		